

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$$\begin{array}{r}
 \text{Divisor } 4 \overline{) 10} \\
 \underline{- 8} \\
 2
 \end{array}$$

$2 \leftarrow \text{Quotient}$
 $10 \leftarrow \text{Dividend}$
 $2 \leftarrow \text{Remainder}$

$$\text{Remainder} = \text{Dividend} - (\text{Divisor} \times \text{Quotient})$$

Highest multiple of
divisor $\leq$ dividend

Example -:

$$-60 \div 9$$

$$\text{Rem} = -60 - \{ \text{of } 9 \leq -60 \}$$

greatest multiple

$$= -60 - (-63)$$

$$= -60 + 63$$

$$= 3$$

C++, C#, JS, Java = $-6 + 9 = 3$
python = 3

```
res = a % b  
if (res < 0)  
    res = res + b
```

* Range:-

N/A	min = 0	$[0, 9]$
$(-\infty, \infty)$	max = 9	range.

* Why Modulo:-

Modulo operator allows us to set the result in specific range.

Use cases:-

- 1) Data distribution - consistent Hashing
- 2) Hashmap / Hashsets - Hashing
- 3) Cryptography
- 4) Sharding in DBMS

* Modular Arithmetic :-

$$(a + b) \% M = (a \% M + b \% M) \% M$$

$\text{range} [0, M-1]$

$$(a * b) \% M = (a \% M * b \% M) \% M$$

$$(a \% p) \% p = a \% p$$

* Divisible by 3 :-

Sum of all digit is divisible
by 3

$$789 \Rightarrow 7 + 8 + 9 = 16 = 24 \checkmark$$

$$8215 \Rightarrow 8 + 2 + 1 + 5 = 16 \times$$

* Divisible by 4 :-

if last 2 digit divisible by 4

$$25 \underline{24} \checkmark, 139 \underline{60} \checkmark, 44 \underline{838} \times$$

* Divisibility by 8:-

The divisibility rule for 8 states the Number is divisible by 8 if Number formed by it's last 3 digit divisible by 8.

e.g. $\underline{456}$, $\underline{\underline{1248}}$

* Divisibility by 9:-

The divisibility for 9 states that a number is divisible by 9 if the sum of it's digit divisible by 9.

e.g. $2763 = 2 + 7 + 6 + 3 = 18/9$

* Power:-

Example

$$a=3, n=4, p=7$$

$$3^4 \cdot 7 = 81 \cdot 7 = 4$$

$$Tc: O(N)$$

$$Sc: O(1)$$

→ How to handle the overflow problem?

$$\text{ans} = (a \% p + a \% p) \% p$$

$$T.C: O(N)$$

$$S.C: O(1)$$

* Mod of large Number in array:-

$$\text{arr} = 7 \ 8 \ 9 \ 6 \ 2 \quad p = 5$$

$$= 78962 \% 5 = 2$$

$$d = A[i] \% p$$

$$x = x \% p$$

$$\text{term} = (d * x) \% p$$

$$T.C: O(N)$$

$$S.C: O(1)$$